Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Simplify $2^3 \times 2^4$.
2. Express 3^{-2}.
3. Solve $2x + 1 = 5$.
4. Differentiate $x^2 + 2x$.
5. What multiplicative factor corresponds to a 20% decrease?
6. For a geometric sequence $u_0 = 5, q = 2$, calculate u_n.

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Explain the difference between a continuous rate and a discrete multiplicative factor; present an inequality proved by finding the minimum of a difference.

Investigate before formalising

Compare the increments of e^x and the values of the function around 0, with several positive steps. Conjecture a relation between $\dfrac{e^h - 1}{h}$ and 1 when h becomes small.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.